

Project Design Phase-II: Data Flow Diagram & User Stories

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Project Name: Heart Disease Analysis[cite: 26]

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1. Data Flow Diagram (DFD) Specifications

A Data Flow Diagram (DFD) visually maps out how patient clinical data moves through the predictive system, detailing where information originates, how it undergoes transformation, and where it is saved[cite: 26].

DFD Level 0 — Context Diagram

- **External Entities:** Cardiologist, Healthcare Data Analyst[cite: 26]
- **Central System (0.0):** Heart Disease Analysis & Patient Risk Analytics System[cite: 26]
- **Data Store (D1):** Heart Disease Clinical Dataset (CSV) — ~50,000 Patient Records[cite: 26]
- **Data Flows:**
 - *Input:* Raw clinical patient files stream from the data store into the analytics engine[cite: 26].
 - *Interaction:* User selections filter physiological risk profiles from both the Cardiologist and Healthcare Analyst perspectives[cite: 26].
 - *Output:* Interactive risk-stratification dashboards and comprehensive clinical reports return to the healthcare professionals[cite: 26].

DFD Level 1 — Process Breakdown

- **Process 1.0 (Data Cleaning):** Cleans raw clinical parameters (handles missing values, resolves out-of-bound resting blood pressure data, and standardizes heart rates)[cite: 26].
 - **Process 2.0 (Data Preparation):** Encodes categorical health tags (e.g., classifying chest pain severity, mapping fasting blood sugar levels, and grouping patient age brackets)[cite: 26].
 - **Process 3.0 (Tableau Dashboard Development):** Builds out responsive visualization metrics capturing critical patient cardiac health markers[cite: 26].
 - **Process 4.0 (Tableau Story Development):** Creates sequential clinical narratives tracking multi-variable epidemiological trends across patient groups[cite: 26].
 - **Process 5.0 (Flask Web Application Integration):** Embeds the secure interactive analytics visual elements directly into an accessible web interface[cite: 26].
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2. System User Stories

User Type	Functional Requirement (Epic)	User Story ID	User Story / Task Description	Acceptance Criteria	Priority	Release
Cardiologist	Clinical KPI Dashboard[cite: 26]	USN-1[cite: 26]	As a Cardiologist, I want to view key metrics such as total patient counts, high-risk case percentages, and average metrics so I can quickly assess incoming clinical load. [cite: 26]	KPI summary cards display accurate, up-to-date totals for patient cohorts when the app loads. [cite: 26]	High[cite: 26]	Sprint-1[cite: 26]
Cardiologist	Chest Pain Type Classification[cite: 26]	USN-2[cite: 26]	As a Cardiologist, I want to analyze the prevalence of different chest pain types so that I can prioritize urgent diagnostic testing resources.[cite: 26]	Dashboard displays case distribution by chest pain category via a clear, sortable visualization. [cite: 26]	High[cite: 26]	Sprint-1[cite: 26]
Cardiologist	Biometric Risk Breakdown[cite: 26]	USN-3[cite: 26]	As a Cardiologist, I want to see how biometric factors (cholesterol, blood pressure) align across patient records to identify severe	Biometric correlation metrics dynamically update whenever primary health criteria filters change. [cite: 26]	Medium[cite: 26]	Sprint-2[cite: 26]

			risk overlaps. [cite: 26]			
Cardiologist	Diagnostic Test Evaluation[cite: 26]	USN-4[cite: 26]	As a Cardiologist, I want to compare abnormality flags across diagnostic tests (ECG, stress test max heart rates) to recognize subtle early risk patterns. [cite: 26]	Diagnostic charts display clear proportional breakdowns of positive vs. negative indicators across test groups.[cite: 26]	Medium[cite: 26]	Sprint-2[cite: 26]
Healthcare Analyst	Interactive Medical Filters[cite: 26]	USN-5[cite: 26]	As a Healthcare Analyst, I want to filter dashboard patient data by specific age ranges and geographical regions to focus on precise sub-populations. [cite: 26]	Activating any demographic filter refreshes all visible charts instantly without requiring a full page reload.[cite: 26]	High[cite: 26]	Sprint-1[cite: 26]
Healthcare Analyst	Patient Diagnostic Trends[cite: 26]	USN-6[cite: 26]	As a Healthcare Analyst, I want to contrast average risk markers against patient lifestyle choices to pinpoint target areas for preventative outreach programs.[cite: 26]	Comparative views prominently flag patient lifestyle cohorts that fall significantly below or above baseline risk averages. [cite: 26]	High[cite: 26]	Sprint-2[cite: 26]

Healthcare Analyst	Diagnostic Timing Trends[cite: 26]	USN-7[cite: 26]	As a Healthcare Analyst, I want to analyze time trends from initial chest pain presentation to definitive diagnosis to call out potential operational delay risks. [cite: 26]	Clinical processing timeline displays a clear trend direction and highlights bottleneck windows beyond medical guidelines. [cite: 26]	High[cite: 26]	Sprint-2[cite: 26]
Healthcare Analyst	Regional Pattern Analysis[cite: 26]	USN-8[cite: 26]	As a Healthcare Analyst, I want to view patient volumes and high-risk classifications by location to identify regional public health trends. [cite: 26]	Regional chart reflects accurate geographic patient distribution mapping directly back to the database. [cite: 26]	Medium[cite: 26]	Sprint-3[cite: 26]
Healthcare Analyst	Patient Age Group Analysis[cite: 26]	USN-9[cite: 26]	As a Healthcare Analyst, I want to observe heart disease trends by patient age groups to see how physiological vulnerabilities change over time.[cite: 26]	Age bracket charts can be layered or filtered by lifestyle choices and key biometric parameters. [cite: 26]	Low[cite: 26]	Sprint-3[cite: 26]
Healthcare Analyst	Risk Aggregation Tracking[cite: 26]	USN-10[cite: 26]	As a Healthcare Analyst, I want to track how high-risk cases aggregate over seasonal or	Temporal trend graphs show correct temporal data roll-ups aggregated by	High[cite: 26]	Sprint-1[cite: 26]

			yearly timelines to forecast department surge periods. [cite: 26]	week, month, or quarter. [cite: 26]		
Healthcare Analyst	Tableau Story Integration[cite: 26]	USN-11[cite: 26]	As a Healthcare Analyst, I want to present a structured Tableau Story summarizing critical epidemiological insights without building fresh slide presentations. [cite: 26]	Story navigation steps sequentially through key clinical arguments, loading the correct, active data panel at each point. [cite: 26]	Medium[cite: 26]	Sprint-3[cite: 26]
Clinical Director	Executive Health KPI Deck[cite: 26]	USN-12[cite: 26]	As a Clinical Director, I want to monitor overall cardiovascular department health indicators at a glance to steer hospital-wide medical policy efficiently.[cite: 26]	Executive clinical summary deck is positioned prominently at the top of the interface layout upon initial application boot.[cite: 26]	High[cite: 26]	Sprint-1[cite: 26]
Clinical Director	Clinical Outcome Reports[cite: 26]	USN-13[cite: 26]	As a Clinical Director, I want to export summary dashboard findings to easily circulate analytical health data with the hospital's	Document export feature creates a clean printable layout capturing the current filtered	Medium[cite: 26]	Sprint-4[cite: 26]

			medical board. [cite: 26]	clinical view. [cite: 26]		
Clinical Director	Cross-Variable Risk Assessment[cite: 26]	USN-14[cite: 26]	As a Clinical Director, I want to contrast risk trends side-by-side across distinct patient demographics to prioritize funding and resource deployment. [cite: 26]	Comparative view allows separate multi-variable patient demographics to be selected and analyzed in a side-by-side layout. [cite: 26]	Medium[cite: 26]	Sprint-3[cite: 26]
Clinical Director	Treatment Optimization[cite: 26]	USN-15[cite: 26]	As a Clinical Director, I want to identify the most successful clinical path strategies across specific heart risk categories to optimize patient length of stay.[cite: 26]	Outcome strategy ranking lists are sortable by post-treatment survival rate or average hospital care duration.[cite: 26]	Low[cite: 26]	Sprint-4[cite: 26]
Portal User	Flask Web Portal[cite: 26]	USN-16[cite: 26]	As a Portal User, I want to safely access the embedded Tableau sheets using a lightweight web interface without downloading additional local software.[cite: 26]	Secure Flask web frame container loads and displays the analytics interface seamlessly across common hospital web browsers. [cite: 26]	High[cite: 26]	Sprint-1[cite: 26]

Portal User	Clear Interface Navigation[cite: 26]	USN-17[cite: 26]	As a Portal User, I want a clean, well-spaced dashboard interface so I can quickly shift between different patient diagnostic views during clinical rounds. [cite: 26]	Menu toggles allow quick hopping across dashboard views (Clinical KPIs, Biometrics, Lifestyle, Diagnostic Tests).[cite: 26]	Medium[cite: 26]	Sprint-2[cite: 26]
Portal User	Unified View Reset[cite: 26]	USN-18[cite: 26]	As a Portal User, I want to reset all analytical data filters back to baseline default values with a single click to begin a new case review.[cite: 26]	A dedicated "Reset Filters" action instantly drops all specific demographic overrides and re-displays standard defaults.[cite: 26]	Low[cite: 26]	Sprint-2[cite: 26]